# Coding&Solution

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| **Date** | 5July2026 |
| **TeamID** | |
| **ProjectName** | CreditCardApprovalPrediction |
| **Maximum Marks** | 5Marks |

## Coding&Solution

### SolutionSummary

| Field | Details |
|---|---|
| **RepositoryLink/URL** | https://github.com/asmath2031-stack/credit-card-approval-project.git |
| **Programming Language(s)** | Python3.12,HTML5,CSS3 |
| **Framework(s)Used** | Flask,Scikit-learn,Pandas,NumPy |
| **KeyFeatures Implemented** | • User-friendlywebinterfaceforenteringapplicantdetails.<br>• MachineLearningmodeltrainedusingRandomForestClassifier.<br>• PredictswhetheracreditcardapplicationshouldbeApprovedor Rejected.<br>• Real-timepredictionthroughFlaskwebapplication.<br>• DatapreprocessingandfeatureengineeringusingPandasandNumPy.<br>• ResponsivefrontenddesignedwithHTMLandCSS. |
| **Pending/Incomplete Features** | Nopendingfeatures.Allplannedfunctionalities havebeen implemented successfully. |
| **Setup/RunInstructions** | 1. Cloneordownloadtheprojectrepository.<br>2. Createavirtualenvironment:<br>python -m venv .venv<br><br>3. Activatethevirtualenvironment:<br>.\.venv\Scripts\Activate.ps1<br><br>4. Install requiredpackages:<br>pipinstall-rrequirements.txt<br><br>5. Navigateto:<br>5. ProjectDevelopmentPhase\SourceCode<br><br>6. Runtheapplication:<br>python app.py<br><br>7. Openthebrowserandvisit:<br>http://127.0.0.1:5000 |

**CodeQualityChecklist**

| S.No | Criteria | Status(Yes /No) |
|---|---|---|
| 1 | Codeismodularandorganizedintofunctions/classes | Yes |
| 2 | Meaningfulvariableandfunctionnamesareused | Yes |
| 3 | Codeincludescomments/documentationwherenecessary | Yes |
| 4 | Errorhandlingisimplementedforcriticaloperations | Yes |
| 5 | Theapplicationrunswithoutcriticalerrors | Yes |
| 6 | Codeiscommittedtoaversioncontrolrepository | Yes |

**AdditionalNotes/Comments:**

The Credit Card Approval Prediction system is developed using Python, Flask, and Machine Learning. Theapplicationprovidesanintuitivewebinterfaceforuserstoenterapplicantinformation and instantly receive an approval prediction generated by a trained Random Forest model. The project follows a modular architecture, maintains clean coding standards, and can be easily extended with additional machine learning models or integrated with a database in future enhancements.